

> restart; **EJEMPLO 2**

> edo := y'' + 0.447²*y = 0;

$$edo := \frac{d^2}{dx^2} y(x) + 0.199809 y(x) = 0 \quad (1)$$

> cond := y(0) = 1, D(y)(0) = 2;

$$cond := y(0) = 1, D(y)(0) = 2 \quad (2)$$

> sol := dsolve({edo, cond}, y(x));

$$sol := y(x) = \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right) \quad (3)$$

> des := rhs(sol);

$$des := \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right) \quad (4)$$

> vel := diff(des, x);

$$vel := 2 \cos\left(\frac{447}{1000} x\right) - \frac{447}{1000} \sin\left(\frac{447}{1000} x\right) \quad (5)$$

> acel := diff(des, x\$2);

$$acel := -\frac{447}{500} \sin\left(\frac{447}{1000} x\right) - \frac{199809}{1000000} \cos\left(\frac{447}{1000} x\right) \quad (6)$$

> ur := 2.4504;

$$ur := 2.4504 \quad (7)$$

> vmax := 2.05;

$$vmax := 2.05 \quad (8)$$

> amax := 0.916;

$$amax := 0.916 \quad (9)$$

> plot([des, vel, acel, ur, vmax, amax], x = 0..20, y = -5..5, color = [blue, red, green, orange, orange, orange], legend = ["u(t)", "u'(t)", "u''(t)", "ur", "vmax", "amax"], title = "Vibración libre sin amortiguamiento", gridlines = true);

Vibración libre sin amortiguamiento

